



MANAGEMENT ACCOUNTING

8th EDITION

BY

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PowerPoint Presentation by
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4 ACTIVITY-BASED PRODUCT COSTING

LEARNING OBJECTIVES

After studying this chapter, you should be able to:



LEARNING OBJECTIVES

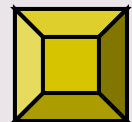
1. Discuss the importance of unit costs.
2. Describe functional-based costing approaches.
3. Tell why functional-based costing approaches may produce distorted costs.

Continued

LEARNING OBJECTIVES

4. Explain how an activity-based costing system works for product costing.
5. Explain how the number of activity rates can be reduced.

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT: SpringBanc

What are product costs?

QUESTIONS TO THINK ABOUT: SpringBanc

What role do product costs play
in bids?

QUESTIONS TO THINK ABOUT: SpringBanc

What is meant by a traditional functional-based costing system?
Why might it cause distortions in product costs?

QUESTIONS TO THINK ABOUT: SpringBanc

Assuming SpringBanc's problems are founded in the way costs are assigned to products, what can SpringBanc do to solve the problem?

LEARNING OBJECTIVE

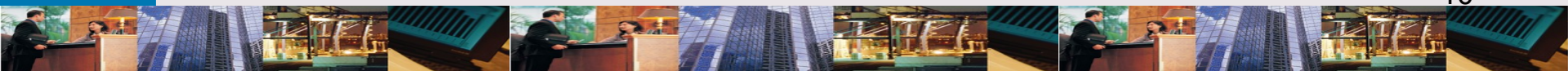
1

Discuss the importance of unit costs.

UNIT COST: Definition

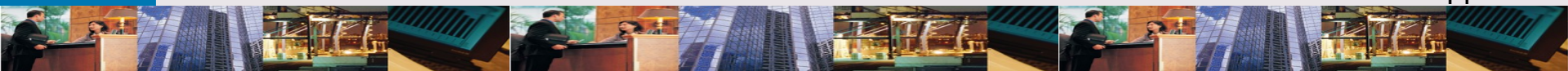
Unit cost is the total cost associated with the units produced divided by the # of units produced.

(Total cost) / (# units produced)



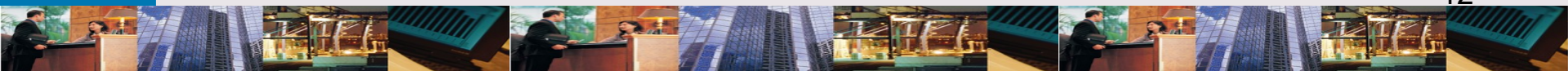
What is meant by “total cost”?

Total cost refers to production costs: direct materials, direct labor, manufacturing overhead.



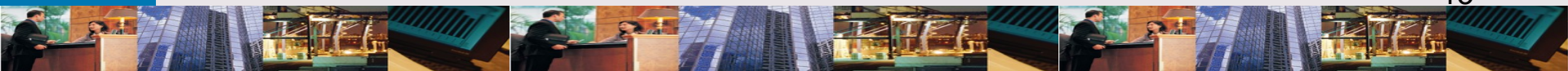
How do we measure the costs
to be assigned?

Production costs we assign may
be **actual** or **estimated** costs.



How do we assign costs to a product?

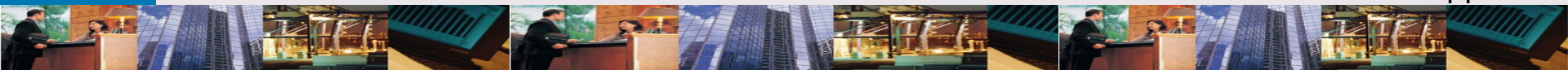
Production costs are assigned by 1 of 2 methods: **functional-based** or **activity-based** costing.



UNIT COSTS

✧ Affect

- ✧ Bids submitted for special products
- ✧ Development, introduction of new products
- ✧ Decisions to
 - ✧ Make or buy a product or service
 - ✧ Accept or reject a special order
 - ✧ Keep or drop a product or service



UNIT COST INFORMATION

- ✧ May be produced by
 - ✧ Actual direct materials, direct labor, overhead
 - ✧ Normal costing
 - ✧ Uses actual costs for direct materials & direct labor but overhead is assigned at **predetermined rates**

$$\text{Predetermined rate} = \frac{\text{Budgeted cost}}{\text{Estimated activity usage}}$$

LEARNING OBJECTIVE

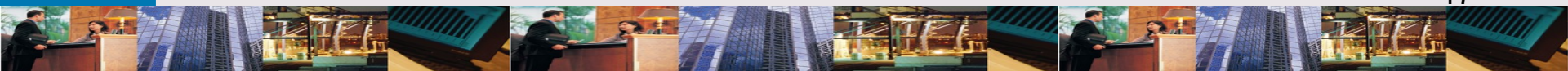
2

Describe functional-based costing approaches.

How are functional-based costs assigned?

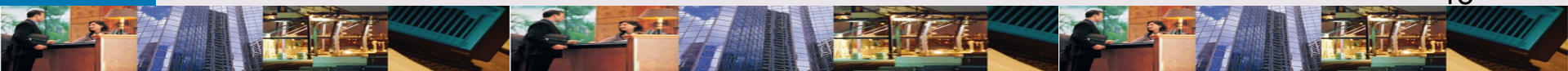
Direct materials & labor are assigned by **direct tracing**.

Overhead is assigned by **driver tracing & allocation**.



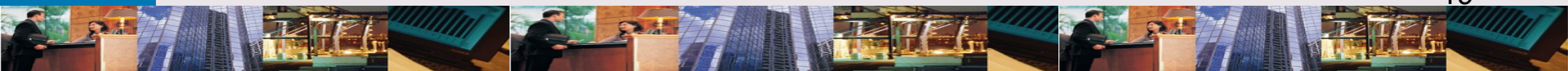
UNIT-LEVEL DRIVERS

- ✧ Used to assign manufacturing overhead by
 - ✧ Units produced
 - ✧ Direct labor hours
 - ✧ Direct labor dollars
 - ✧ Machine hours
 - ✧ Direct material dollars



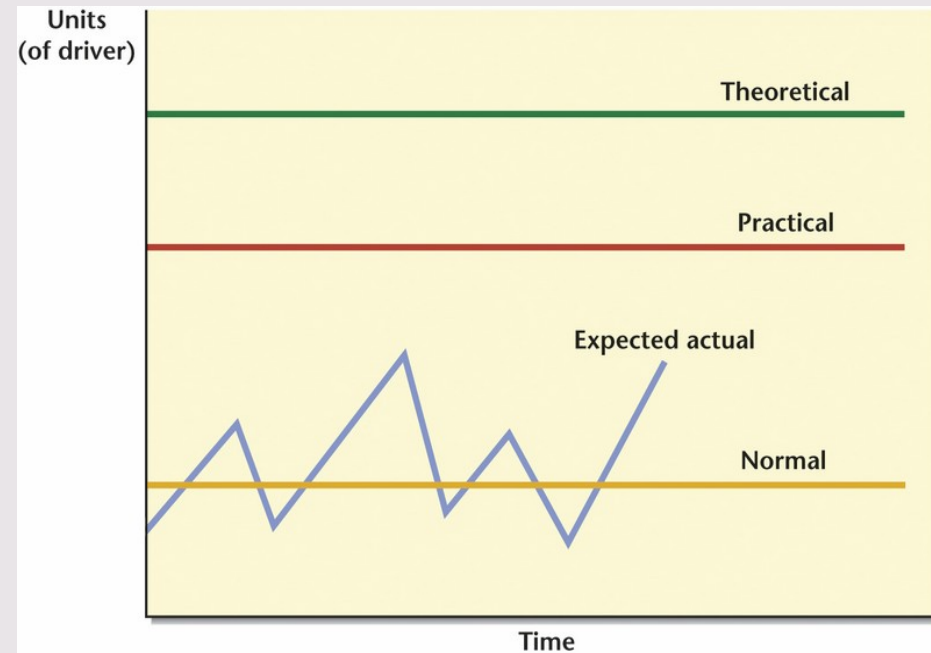
CAPACITY LEVELS: Definition

- ✧ Expected activity capacity is output for coming year
- ✧ Normal activity capacity is average activity experienced over long term
- ✧ Theoretical activity capacity is absolute maximum that can be achieved in perfect world
- ✧ Practical activity capacity is maximum that can be achieved under efficient operation



What does the relationship of capacity levels look like?

EXHIBIT 4-1



PLANTWIDE RATE:

An Example

Budgeted overhead	\$360,000
Expected activity (in DLH)	100,000
Actual activity (in DLH)	100,000
Actual overhead	\$380,000

Predetermined rate = Budgeted cost / Estimated activity usage

$$= \$360,000 / 100,000$$

$$= \text{\textbf{\$3.60 per DLH}}$$

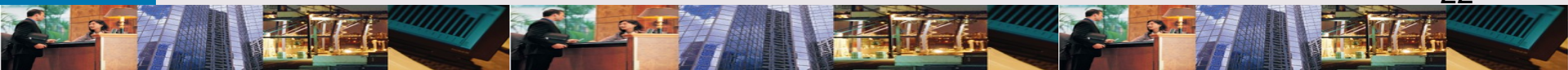
Applied overhead = Overhead rate x Actual activity

$$= \$3.60 \times 100,000$$

$$= \text{\textbf{\$360,000}}$$

What if applied overhead (\$360,000) differs from actual overhead (\$380,000)?

Underapplied (**overapplied**) overhead is a variance that is **added to** (**subtracted from**) cost of goods sold.



PLANTWIDE RATE:

2 Products

Similar products use different amounts of labor, so that \$360,000 overhead is allocated differently.

	Cordless	Regular
Prime costs	\$ 78,000	\$ 738,000
Overhead costs:		
$\$3.60 \times 10,000$	36,000	—
$\$3.60 \times 90,000$	—	<u>324,000</u>
Total manufacturing costs	\$114,000	\$1,062,000
Units produced	$\div 10,000$	$\div 100,000$
Unit cost (total costs/units)	<u>\$ 11.40</u>	<u>\$ 10.62</u>

Exhibit 4-3 Unit-Cost Computation: Plantwide Rate—BelRing Springdale Plant

EXHIBIT 4-3

DEPARTMENTAL RATE MODEL

Products produced in different departments use different drivers to assign overhead costs.

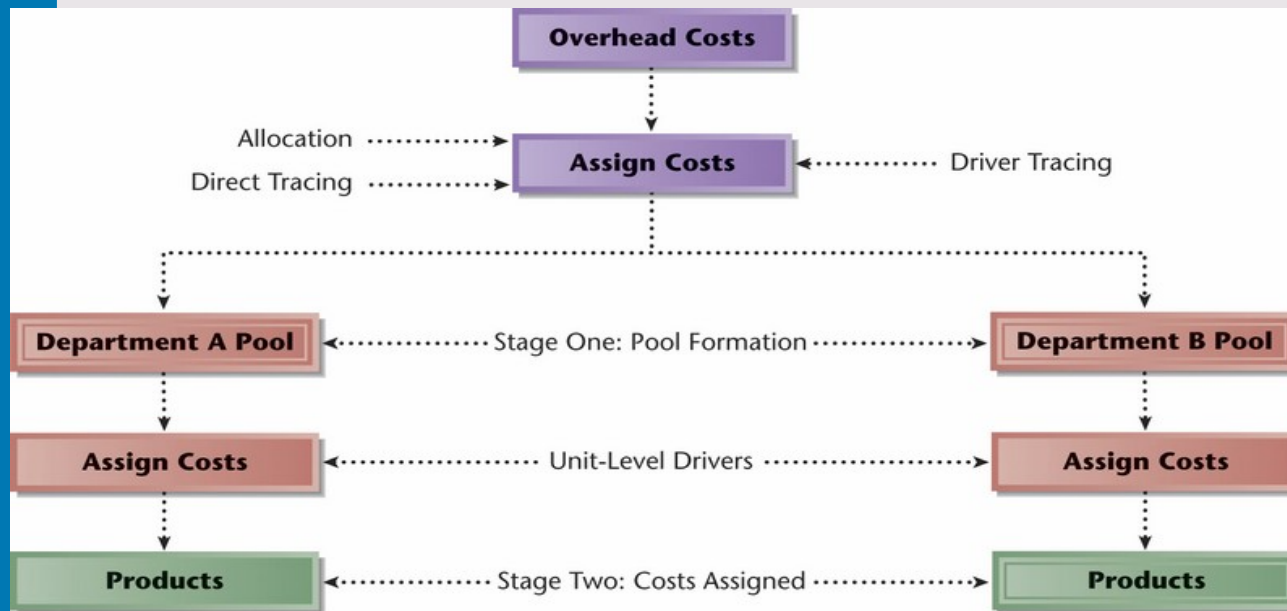


EXHIBIT 4-4

DEPARTMENTAL RATE:

2 Products

When different drivers are used in 2 or more departments, driver tracing assigns \$360,000 overhead differently.

	Fabrication	Assembly
Budgeted overhead	<u>\$252,000</u>	<u>\$108,000</u>
Expected and actual usage (direct labor hours):		
Cordless	7,000	3,000
Regular	<u>13,000</u>	<u>77,000</u>
	<u>20,000</u>	<u>80,000</u>
Expected and actual usage (machine hours):		
Cordless	4,000	1,000
Regular	<u>36,000</u>	<u>9,000</u>
	<u>40,000</u>	<u>10,000</u>

Exhibit 4-5 Departmental Data—BelRing Springdale Plant

EXHIBIT 4.5

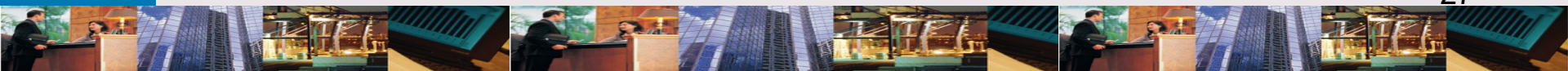
LEARNING OBJECTIVE

3

Tell why functional-based costing approaches may produce distorted costs.

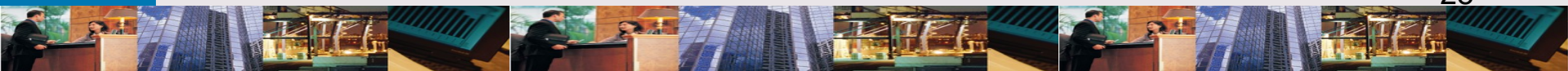
How does a company know if its functional-based costing is producing distorted costs?

There are many symptoms of distorted costs, such as profit margins that are difficult to explain.



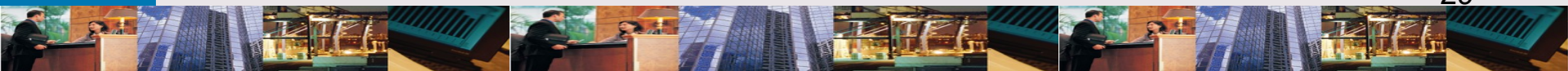
CREATING DISTORTIONS

When a unit-level approach is used to assign non-unit-level costs ((overhead)), cost information can be distorted.



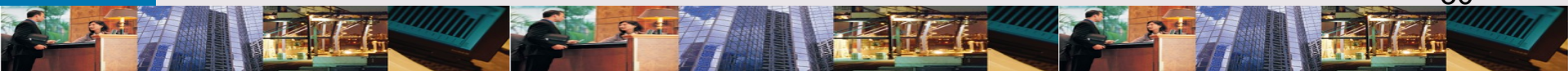
NON-UNIT-LEVEL ACTIVITY DRIVER: Definition

Factors that measure consumption of non-unit-level activities by product & can distort product costs.



PRODUCT DIVERSITY: Definition

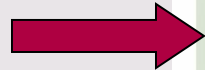
Products consume overhead activities in systematically different proportions.



COST DISTORTIONS:

Plantwide Cost Assignment

Proportional application assigns **9 times** as much overhead to Regular phones.



Activity Usage Measures			
	Cordless	Regular	Total
Units produced per year	10,000	100,000	110,000
Prime costs	\$78,000	\$738,000	\$816,000
Direct labor hours	10,000	90,000	100,000
Machine hours	5,000	45,000	50,000
Production runs	20	10	30
Number of moves	60	30	90

Activity Cost Data (Overhead Activities)	
Activity	Activity Cost
Setups	\$120,000
Material handling	60,000
Machining	100,000
Testing	80,000
Total	<u>\$360,000</u>

Exhibit 4-8 Product-Costing Data—BelRing Springdale Plant

Should we assume that regular phones will use **9 times more** overhead costs than cordless phones?

EXHIBIT 4-8

COMPARING UNIT COSTS

Activity-based costing
is more realistic than
functional-based
costing.

	Cordless	Regular	Source
Plantwide rate	\$11.40	\$738,000	Exhibit 4-3
Departmental rate	10.73	10.69	Exhibit 4-6
Activity rate	21.60	9.60	Exhibit 4-10

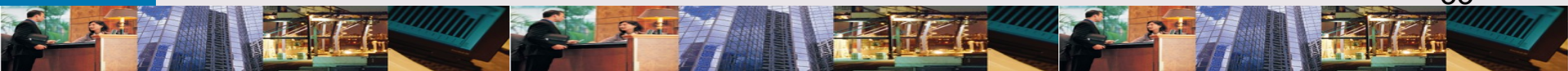
Exhibit 4-11 Comparison of Unit Costs

EXHIBIT 4-11

LEARNING OBJECTIVE

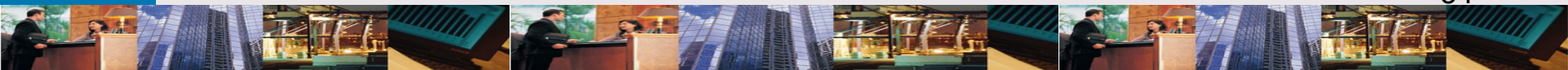
4

Explain how an activity-based costing system works for product costing.



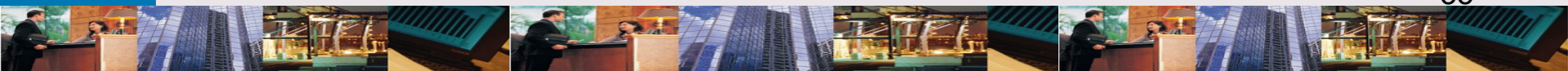
How do you identify activities
& their attributes?

Use a key set of interview
questions and record the
answers.



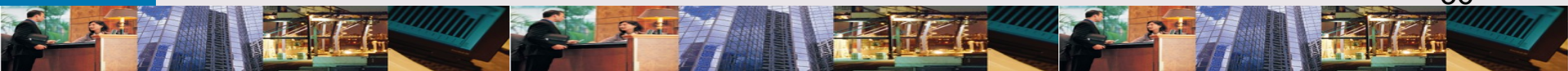
KEY QUESTIONS

- ✧ How many employees are in your department?
- ✧ What do they do?
- ✧ Do customers outside your department use any equipment?
- ✧ What resources are used by each activity?
- ✧ What are the outputs of each activity?
- ✧ Who or what uses the activity output?
- ✧ How much time do workers spend on each activity?
By equipment?



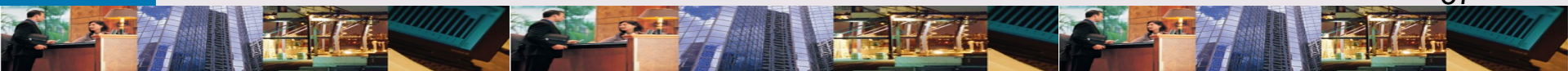
RESOURCE DRIVER: Definition

Factors that measure the consumption of resources by activities.



ASSIGNING COSTS TO **ACTIVITIES**: Step 1

Activity	% of Time on Each Activity	
	Supervisor	Clerks
Supervising employees	100%	0%
Processing transactions	0	40
Preparing statements	0	30
Answering questions	0	30



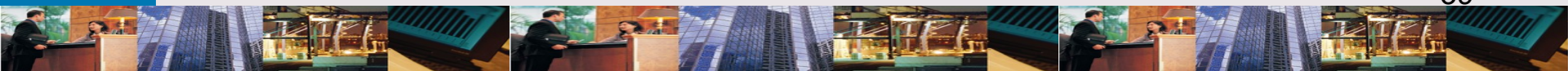
ASSIGNING COSTS TO **ACTIVITIES**: Step 2

If the supervisor's salary is \$50,000 and each of 5 clerks earn \$30,000, costs would be:

Activity	Supervisor	Clerks
Supervising employees	\$50,000	---
Processing transactions	---	\$60,000
Preparing statements	---	\$45,000
Answering questions	---	\$45,000

How do you assign activity costs?

Activity costs are assigned to products on the basis of usage.



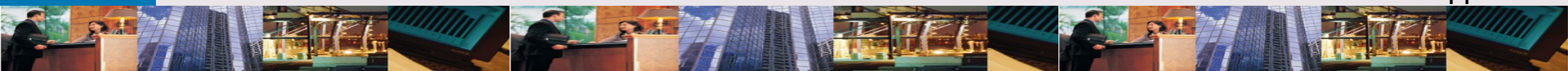
ASSIGNING COSTS TO PRODUCTS: Identify Costs

EXHIBIT 4-14

Activity Costs	
Activity	Credit Card
Supervising employees	\$ 75,000
Processing transactions	100,000
Preparing statements	79,500
Answering questions	69,900
Providing automatic tellers	250,000

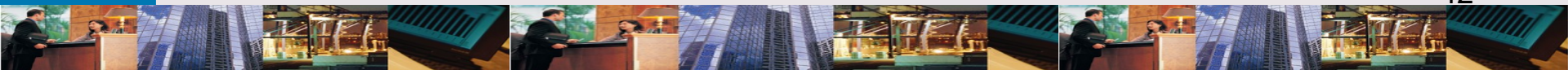
ASSIGNING COSTS TO PRODUCTS: Assign Costs

Activity	Classic Gold	Gold	Platinum	Total
# Cards	5,000	3,000	2,000	10,000
Transactions processed	600,000	300,000	100,000	1,000,000
# Statements	60,000	36,000	24,000	120,000
# Calls	10,000	12,000	8,000	30,000
# Teller transactions	15,000	3,000	2,000	20,000



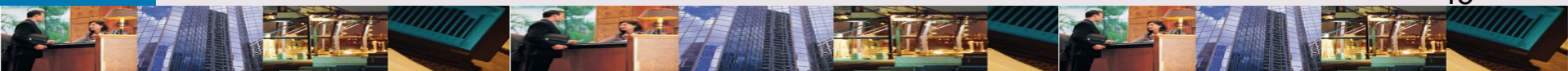
CLASSIFICATION OF ACTIVITIES

Product costs can be assigned at a) unit level, 2) batch level, 3) product level (*as in this example*) or 4) facility level.



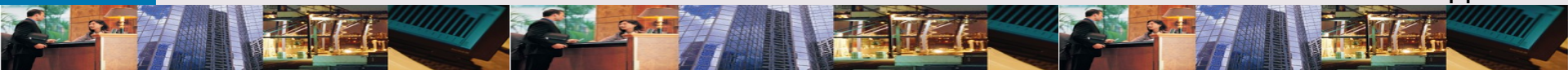
LEARNING OBJECTIVE

5 Explain how the number of activity rates can be reduced.



ACTIVITY BASED COST SYSTEM

- ✧ Reduce size, complexity by reducing rates
 - ✧ Using consumption ratios
 - ✧ By approximating ABC



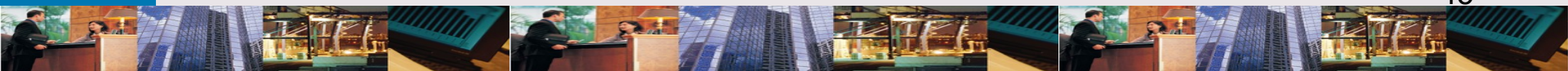
COMPARING SYSTEMS

✧ Functional-based system

- ✧ Overhead uses only unit-based activity drivers or
- ✧ Splits overhead into fixed and variable and allocates overhead based on the appropriate unit-level rate

✧ Activity-based system

- ✧ Improves product costing by
 - ✧ Looking at cause & effect
 - ✧ Recognizing that some fixed overhead varies in proportion to changes other than production volume



CHAPTER 4

THE END